KAUSTUBH KANTILAL ADSUL
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OBJECTIVE

Ph.D. candidate in Mechanical Engineering with extensive research and industry experience in advanced thermal management and data-center cooling technologies. Experienced in two-phase and single-phase immersion cooling, indirect evaporative cooling, liquid cold plates, thermal interface materials, and thermal reliability testing. Strong expertise in experimental thermal characterization, CFD modeling and validation using ANSYS Fluent, Flownex and IcePak, thermal instrumentation, data acquisition, and system optimization. Proven experience leading industry-collaborative projects, including on-site deployment and commissioning of a 10 MW two-phase immersion cooling system. Seeking full-time roles in data center thermal management, starting after Ph.D. completion in Dec 2026.

WORK EXPERIENCE

Marathon Digital Holdings, Project Engineer, Two-Phase Immersion Cooling, June 2025 – August 2025.

- Led on-site deployment and installation of two-phase immersion cooling (2PIC) systems.
- Managed project planning, resources, deliverables, and deployment timelines.
- Coordinated MEP contractors and cross-functional teams for system integration.
- Managed miner energization, SCADA implementation, sensor instrumentation, and data acquisition.
- Analyzed sensor data to support thermal-performance optimization and miner overclocking.
- Validated thermal interfaces, dielectric fluid levels, pressure integrity, and system alignment.
- Executed commissioning, leak testing, fluid circulation verification, and performance validation.
- Led on-site troubleshooting and technical coordination to ensure reliable system deployment.

EMNSPC Lab, University of Texas at Arlington, Graduate Research Assistant, January 2020 – Present.

- Proficient in simulation and design tools including ANSYS IcePak, Fluent, and SolidWorks for CFD modeling and thermal analysis.
- Led an experimental project on Indirect Evaporative Cooling in partnership with MESTEX.
- Led CFD optimization studies for single-phase immersion cooling and contributed to associated experimental validation with LiquidStack.
- Executed component- and server-level experimental testing for two-phase immersion cooling in collaboration with Chemours.
- Initiated and led the development of a cost-effective in-house Thermal Interface Material (TIM) Tester under $20K (market value: $100K)
- Collaborated on multiple experimental thermal projects with industry partners including Google, ERG Aerospace, and LiquidStack focused on single-phase immersion and two-phase cold plate cooling systems.

EDUCATION

Ph.D. in Mechanical Engineering, *University of Texas at Arlington* *Jan 2022 – Present*
(Expected completion: Dec 2026)
Thesis Topic "*Experimental and Numerical Analysis to Quantify the Effect of Secondary air on Data Center Primary Air using Indirect Evaporative Cooling."*

M.S. in Mechanical Engineering, *University of Texas at Arlington* *Aug 2019 – May 2021*
Thesis Topic *"Thermal Design Analysis of Server Chassis Manifolds for Liquid Cooled Servers using CFD"*

B.E. in Mechanical Engineering, *University of Pune* *Jun 2013 – May 2017*
Project *"Automatic Tire Inflation System using Nitrogen"*

SKILLS

Thermal Testing & Measurement: Thermocouples, RTDs, DAQ (Agilent), Environmental chambers, CDUs (L2L/L2A), Thermal Test Vehicles, Cold plates, Belimo Valves, Chillers, Dry Coolers, Flow & pressure sensors. | **CAD & CAE Software:**
SolidWorks, AutoCAD, ANSYS Workbench, ANSYS IcePak, ANSYS Fluent, Flownex | **Controls, Data & Software:** MATLAB, Project Planner, Microsoft Office. | **Manufacturing & Operations:** Test plan creation, Root-cause analysis, Quality control, SOP development | **Certification:** Certified Mission Critical Operator (CMCO)

PROJECTS

Two-Phase Immersion Cooling

- Led the end-to-end deployment of a 10 MW two-phase immersion cooling site
- Led project planning and coordination, aligning stakeholders on resources, deliverables, and execution timelines.
- Managed energization of miners within tanks, SCADA implementation, sensor wiring, and sensor data analysis leading to overclocking of miners.
- Provided full on-site supervision, troubleshooting, and coordination with cross-functional teams to ensure timely and reliable deployment.

Indirect Evaporative Cooling, Mestex Inc.

- Led experimental characterization of an Indirect Evaporative Cooling (IEC) system for data-center cooling under varying airflow, water flow rate, and nozzle-height conditions.
- Analyzed thermal performance, heat/mass transfer, pressure drop, and spray distribution, using psychrometric analysis and experimental performance metrics.
- Developed and validated CFD models of six-nozzle spray distribution against experimental heat-map data to evaluate wetting uniformity and optimize cooling performance.

Single-Phase Immersion Cooling, LiquidStack

- Collaborated with LiquidStack to design and build a single-phase immersion cooling tank, conducting experimental thermal characterization of multiple heat-sink designs.
- Developed CFD models in ANSYS Fluent to evaluate heat-transfer and fluid-flow performance, validating simulation results against

experimental data.
- Performed heat-sink design optimization using CFD, manufactured the optimized design, and experimentally tested its thermal performance for validation.

Reliability Thermal Interface Material Tester
- Designed and developed an in-house Thermal Interface Material (TIM) tester for characterizing thermal performance under controlled pressure, interface area, and material conditions.
- Integrated cartridge heaters, copper hot blocks, RTDs, load cells, and LVDT instrumentation with a compression testing system for thermal and mechanical characterization.
- Characterized TIMs to determine thermal resistance and apparent thermal conductivity, enabling evaluation of material performance across varying test conditions.

Reliability Study, OCP (Shell, ExxonMobil, Valvoline)
- Conducted coupon preparation, pre-weighing, test-fluid preparation, and controlled aging experiments for OCP-related material evaluation.
- Managed aging batches and post-processing, including coupon cleaning, imaging, and post-aging weight characterization.
- Performed FTIR, TAN, conductivity, ICP, SEM, and XPS analysis to characterize material and fluid changes after aging.

PUBLICATIONS

In Preparation
- **K. Adsul**, L. Reddy, U. Chinomso, R. Paranjpee, P. Satpute, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental characterization of nozzle height on secondary-air heat transfer in a crossflow indirect evaporative cooling test facility under variable operating conditions," *Journal manuscript in preparation.*
- **K. Adsul**, T. Lambe, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental mapping of collected-water distribution in a crossflow indirect evaporative cooling test facility under variable nozzle height, water flow, and fan speed," *Journal manuscript in preparation.*

Conference Papers
- M. Tangri, K. Rahul, **K. Adsul**, M. Emami, Y. Fulpagare, D. Agonafer, "A comparative study of deep learning architectures for automated classification of particulate contaminants on electronic packaging surfaces," *ASME InterPACK 2026.*
- M. Tangri, **K. Adsul**, "Toward compute-efficient, boundary-condition-aware machine learning surrogates for tile-based IC thermal prediction," *IEEE ITherm 2026.*
- P. Jayaramu, M. Emami, V. Talari, M. R. Islam, **K. Adsul**, R. K. Suthar, et al., "Thermo-hydraulic performance of targeted flow in aluminum and copper heat sinks for immersion cooling applications," *IEEE ITherm 2025, Dallas, TX, pp. 1–9. doi: 10.1109/ITherm55376.2025.11235582*
- M. Emami, R. K. Suthar, P. Bansode, M. Tangri, U. Rahangdale, S. Barua, D. Bhattarai, S. Amininasab, **K. K. Adsul**, D. Agonafer, "Investigation of the thermomechanical behavior of ASTRA M77 substrate core under single-phase liquid immersion cooling," *ASME InterPACK 2025, IPACK2025-169286. doi: 10.1115/IPACK2025-169286*
- S. Saini, **K. K. Adsul**, P. Shahi, A. Niazmand, P. Bansode, D. Agonafer, "CFD modeling of the distribution of airborne particulate contaminants inside data center hardware," *ASME InterPACK 2020, IPACK2020-2590. doi: 10.1115/IPACK2020-2590*

Posters
- **K. Adsul**, T. Lambe, U. Chinomso, R. Paranjpee, P. Satpute, T. Mistry, Y. Fulpagare, D. Agonafer, "Experimental characterization of nozzle spray distribution at varying heights on a crossflow heat exchanger," *InterPACK 2026.*
- **K. Adsul**, Y. Fulpagare, D. Agonafer, "Experimental mapping of spray distribution under variable nozzle height and operating conditions in a crossflow heat exchanger," *ES2 2026.*
- **K. Adsul**, S. Saini, P. Shahi, A. Niazmand, P. Bansode, D. Agonafer, "CFD modeling of the distribution of airborne particulate contaminants inside data center hardware," *ASME InterPACK 2020.*

EXTRA CURRICULAR ACTIVITIES
- Volunteer at Hospice.
- Participated in Society of Automotive Engineers student competitions; served as team lead and contributed to achieving a top 25% national ranking.